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From Signals to Distress: A Time-Aware Machine Learning Framework for ST Classification in China’s Capital Market

Abstract

This study presents a time-aware machine learning framework for predicting financial distress through ST classification in China’s capital market. By modeling financial signals as a temporal sequence rather than static observations, we incorporate lagged financial indicators to capture early signs of deterioration. The framework integrates traditional classifiers with interpretable techniques such as SHAP to reveal the evolving importance of financial features over time. Empirical analysis demonstrates that incorporating temporal dynamics enhances predictive performance and model transparency. The findings suggest that financial distress is better understood as a cumulative process, where historical signals offer critical foresight. This research provides methodological and practical insights for early warning systems, supporting more proactive risk management in emerging markets.

1 Introduction

In recent years, the demand for early identification of corporate financial distress has intensified, particularly within complex and fast-moving markets such as China’s A-share system. Traditional risk modeling frameworks often depend on static, single-period financial indicators. However, substantial empirical evidence suggests that financial deterioration typically unfolds over time—preceded by gradual shifts in key financial metrics such as profitability, leverage, and structural soundness (Kothari et al., 2000).

To address this limitation, the present study constructs a dynamic financial distress prediction model that incorporates both contemporaneous and lagged financial indicators, capturing time points at t , $t-1$, and $t-2$. Specifically, the model leverages **eleven key financial ratios** drawn from five fundamental dimensions: solvency (e.g., TL/TA, CA/CL, TL/TSE), profitability (e.g., NP/SR, NP/ASE), operational efficiency (e.g., SR/ACA, MBC/AI), business growth capacity (e.g., $TA(t)/TA(t-1)$, $NP(t)/NP(t-1)$), and structural soundness (e.g., FA/TA, SE/FA). This temporally extended feature set allows for a richer, time-aware representation of financial health and deterioration (Lee, 2023).

Focusing on data from 2019—a year characterized by increased macroeconomic pressure and escalating U.S.–China trade tensions—this study investigates firms that were subsequently designated as “Special Treatment” (ST) under China’s regulatory framework (Liu and Woo, 2018). The ST designation serves as a tangible proxy for financial distress, enabling supervised learning techniques to model the early-warning characteristics preceding formal classification.

By integrating lag-sensitive financial indicators with machine learning classifiers and SHAP-based interpretability tools, this study not only identifies which financial signals are most predictive, but also *when* these signals become materially relevant. In doing so, it provides a dynamic, interpretable, and practically actionable early-warning system for use by investors, regulators, and corporate decision-makers.

2 Literature review

2.1 From Static Models to Dynamic Financial Distress Prediction

Financial distress rarely occurs instantaneously. Instead, it tends to emerge through gradual and compounding deterioration in a firm’s financial structure, profitability, and solvency. Early approaches to distress prediction framed the problem as a static classification task, relying on financial ratios derived from a single time point to assess default risk (Altman, 1968). Such models often ignored the temporal dependencies and transitional signals embedded in financial statements. More recent work highlights how shifts in reporting patterns across time can significantly impact the predictive reliability of accounting ratios (Beaver, Correia and McNichols, 2012). Corporate decline typically unfolds in stages: firms may first experience weakened operational margins, followed by liquidity pressure and eventually breach of regulatory thresholds. The cumulative and path-dependent nature of these signals underscores the limitations of snapshot-based models, prompting calls for temporally structured frameworks (Ohlson, 1980).

2.2 The Role of Lagged Financial Indicators in Early-Warning Systems

Forecasting models that fail to incorporate lagged features may overlook early symptoms of financial instability. Lagged financial ratios—such as prior-period debt ratios or earnings performance—embed latent indicators that reflect the early emergence of distress long before it becomes publicly observable. Time-aware models improve foresight and allow for earlier intervention, particularly in regulatory and investment contexts (Shumway, 2001).

Multi-period frameworks offer additional advantages, distinguishing between short-term noise and long-term decline. Firms often exhibit delayed responses to macroeconomic shocks, which manifest through gradual erosion of capital adequacy and cash flow health (Tinoco and Wilson, 2013). Furthermore, recent research highlights the value of lag structures in identifying regional economic vulnerabilities and institution-level frictions that may contribute to systemic instability (Rodríguez-Pose and Ketterer, 2020).

2.3 Machine Learning and SHAP-Based Interpretability in Financial Risk Assessment

Advances in machine learning have reshaped financial distress modeling by introducing algorithms capable of learning complex, nonlinear relationships from large datasets. Models such as random forests, support vector machines, and gradient boosting machines have consistently outperformed traditional statistical classifiers in predictive accuracy (Geng, Bose and Chen, 2015). These improvements are especially valuable when dealing with high-dimensional financial signals across multiple timeframes.

However, the opaque nature of most machine learning models has raised concerns about interpretability, particularly among regulators and institutional decision-makers. Explainable AI techniques have gained traction as essential tools to bridge this transparency gap. SHAP (SHapley Additive exPlanations) has emerged as a leading framework, offering consistent and theoretically grounded estimates of each feature’s contribution to the model output (Lundberg and Lee, 2017).

Tree-based recursive architectures have proven effective not only for prediction but also for continuous adaptation to new financial data. These models support incremental updates, real-time retraining, and modular interpretability in dynamic environments (Podhorská et al., 2020).

2.4 Special Treatment (ST) Firms in the Chinese A-share Market

In China’s capital markets, the “Special Treatment” (ST) label signals heightened financial risk and regulatory attention. Firms receive this designation after consecutive years of losses, severe operational anomalies, or negative audit outcomes. Studies show that ST events frequently cluster during macroeconomic slowdowns, suggesting that they serve as a proxy for systemic

financial stress (Zhao, Chen and Zhang, 2019). Beyond financial metrics, behavioral and market-based indicators are also relevant in ST designation. Abnormal accruals are often associated with attempts to obscure deteriorating fundamentals and can precede formal classification events (Yang and Jiang, 2015). Moreover, empirical evidence indicates that ST announcements trigger immediate and sharply negative price adjustments in equity markets, reflecting strong investor sensitivity to perceived distress (Christiawan, Kwistianus, et al., 2023).

2.5 Summary of Literature Gap

Although prior research has enriched the theoretical foundation of financial distress prediction, significant methodological gaps remain. Many models still rely on static financial snapshots, overlooking the rich temporal dynamics embedded in firms' financial trajectories. Few frameworks combine multi-period lagged variables with explainable machine learning algorithms in the specific context of ST firms in China. There is growing demand for predictive systems that are not only statistically robust but also interpretable, scalable, and temporally adaptive. A modeling strategy that integrates lag-sensitive features with SHAP-based interpretability tools offers a promising pathway for advancing early-warning systems in complex financial environments (Cong et al., 2021).

3 Hypothesis Development

Financial distress within firms is often a cumulative process, not an abrupt event. Early signs of financial instability frequently manifest in past financial periods, such as declining profitability, increasing leverage, or weakening cash flows. These lagging signals may precede formal regulatory intervention or market recognition, making them critical inputs for predictive modeling. Empirical studies have confirmed that incorporating lagged financial features significantly improves the identification of at-risk firms in both developed and emerging markets. The rationale is that temporal patterns reflect the underlying evolution of financial health more effectively than static snapshots. Therefore, the hypothesis is proposed:

H1: Lagged financial indicators from previous periods significantly enhance the accuracy of ST company prediction.

Financial deterioration does not follow a uniform path across all firms. Some companies experience immediate and sharp downturns, while others show slow and subtle declines over several quarters. Capturing this trajectory requires a modeling approach that can differentiate between short-term volatility and structural decline. Dynamic modeling frameworks allow for the recognition of sequential risk signals and improve the model's sensitivity to early-stage distress. By examining financial variables over t , $t-1$, and $t-2$, models can better contextualize present risk in light of past performance. Therefore, the hypothesis is proposed:

H2: Incorporating dynamic, multi-period financial features improves the early detection of financial distress compared to single-period models.

While predictive accuracy is crucial, stakeholders in financial markets—such as regulators, investors, and credit analysts—also require transparency in how models arrive at their decisions. The rise of explainable AI has addressed this challenge, with SHAP offering a method to quantify the importance and timing of financial indicators in high-dimensional models. By applying SHAP values, researchers can identify which lagged indicators carry the most weight and how their influence shifts across time. This interpretability supports more informed decision-making and offers insights into which metrics function as early-warning signals versus reactive indicators. Therefore, the hypothesis is proposed:

H3: SHAP-based interpretability reveals the differential temporal importance of financial indicators in predicting ST classification.

4 Data preparation

4.1 Sample Selection and Data Source

This study focuses on all listed firms in China’s A-share market in 2019. From this population, 51 firms labeled as “Special Treatment” (ST) were identified and selected as the target group. For these firms, 20 quarters of financial data were collected from the first quarter of 2015 to the fourth quarter of 2019 using the CSMAR database ¹. For comparison purposes, financial data from the same period were also gathered for all non-ST firms, excluding any that were labeled ST in 2019. A total of 27 financial indicators were computed for both ST and non-ST firms, covering five key financial dimensions. This process yielded two initial datasets: one for ST firms and one for non-ST firms.

Each firm in both datasets was assigned a binary label (ST = 1; non-ST = 0), and the two datasets were then merged into a unified sample (Garrido, Verbeke and Bravo, 2018). Prior to modeling, the merged data were cleaned by addressing data quality issues—such as replacing infinite values with missing values and imputing missing values using column means.

Table 1: Financial Indicators Classification and Calculation Formulas

Types	Symbols	Formulae for Calculation
Solvency (3 indicators)	TL/TA	total liabilities / total assets
	CA/CL	current assets / current liabilities
	TL/TSE	total liabilities / total shareholders’ equity
Profitability (2 indicators)	NP/SR	net profit / sales revenue
	NP/ASE	net profit / average shareholders’ equity
Operational Capabilities (2 indicators)	SR/ACA	sales revenue / average current assets
	MBC/AI	main business cost / average inventory
Business Development Capacity (2 indicators)	TA(t)/TA(t-1)	total assets of this year / total assets of last year
	NP(t)/NP(t-1)	net profit of this year / net profit of last year
Structural Soundness (2 indicators)	FA/TA	fixed assets / total assets
	SE/FA	shareholders’ equity / fixed assets

4.2 Aggregation to the Firm Level

Given that each firm originally had 20 quarterly records, we aggregated them into a single record per firm by computing the average of each indicator (Sahel and Vesala, 2001). This aggregation was essential for three reasons:

1. It reduces noise caused by temporary quarterly shocks (e.g., an extreme profit dip in 2016).
2. It avoids confusion between short-term anomalies and long-term structural deterioration, ensuring the model captures overall financial condition rather than isolated fluctuations.
3. It prevents repetitive anomaly detections across quarters for the same firm, avoiding biased outlier counts.

The use of mean values preserves the financial trend of each firm over time and transforms the panel data into a more structured, firm-level dataset suitable for further analysis.

¹CSMAR Database: <https://data.csmar.com/>

4.3 Outlier Detection and Removal

To ensure the robustness of the dataset, we applied the interquartile range (IQR) method to identify and remove outliers from the non-ST firm sample. For each financial indicator, we calculated the first quartile (Q_1) and third quartile (Q_3), and computed the interquartile range as:

$$\text{IQR} = Q_3 - Q_1 \quad (1)$$

Outlier thresholds were then defined as:

$$\text{Lower Bound} = Q_1 - 1.5 \times \text{IQR}, \quad \text{Upper Bound} = Q_3 + 1.5 \times \text{IQR} \quad (2)$$

Firms whose indicator values exceeded these thresholds were flagged as outliers. As shown in Table 2, we recorded the number and percentage of outliers, along with their computed bounds (Vinutha, Poornima and Sagar, 2018). Based on four key indicators— $NP(t)/NP(t-1)$, NP/ASE , FA/TA , and SR/ACA —we removed extreme outliers from the dataset. After cleaning, the number of non-ST firms was reduced from 5763 to 3142, resulting in a total sample size of 3193 (including ST firms).

Table 2: Outlier Statistics for Financial Indicators (Non-ST Firms)

Variable	Outliers	Percentage (%)	Lower Bound	Upper Bound
NP(t)/NP(t-1)	1608	27.90	-0.213	0.574
NP/ASE	1057	18.34	0.004	0.082
FA/TA	886	15.37	0.028	0.326
SR/ACA	878	15.24	0.251	1.047
SE/FA	785	13.62	0.358	0.822
TL/TA	784	13.60	0.178	0.642
CA/CL	487	8.45	0.253	4.112
TA(t)/TA(t-1)	441	7.65	-0.051	0.249
TL/TSE	310	5.38	-0.600	2.906
MBC/AI	63	1.09	-142.499	244.664
NP_SR	26	0.45	-3.393	5.829

4.4 Standardization and KNN-Based Sample Matching

To ensure comparability across different financial features, we standardized the 11 selected core indicators. Given the substantial class imbalance between ST and non-ST firms, a K-nearest neighbors (KNN) matching algorithm was applied to construct a balanced dataset (Guo et al., 2003). Specifically, for each ST firm, three non-ST firms with the most similar financial profiles in the feature space were matched. The similarity between firms was measured using the Euclidean distance:

$$d(x_i, x_j) = \sqrt{\sum_{k=1}^n (x_{ik} - x_{jk})^2} \quad (3)$$

where x_i and x_j denote the i -th and j -th firms in the dataset, and x_{ik} represents the value of the k -th standardized financial indicator for firm x_i . The total number of features used for distance calculation is denoted by n , which equals 11 in this study. All indicators were standardized prior to matching to eliminate the influence of scale differences across variables.

This procedure yielded a final dataset consisting of 51 ST firms and 112 non-ST firms. The resulting sample preserved the characteristics of the ST group while ensuring structural comparability with the control group.

4.5 Lag Feature Construction and SMOTE Oversampling

We returned to the original quarterly data of the 163 selected firms. After standardizing the data, a time variable was added to each record. For every financial indicator, lagged values for one period ($t-1$) and two periods ($t-2$) were generated based on company code. The first two quarters of each firm's data were excluded due to unavailability of lag values, while company code and time fields were retained to support downstream temporal modeling.

To further address the issue of limited minority-class samples, we applied the Synthetic Minority Over-sampling Technique (SMOTE) to generate synthetic ST-like observations. Rather than using a basic linear interpolation between a single sample and its neighbor, we adopted an enhanced SMOTE formulation that incorporates weighted interpolation over k nearest neighbors (Wang et al., 2006). Specifically, for a given minority sample x_i , the synthetic sample x_{new} is generated as follows:

$$x_{\text{new}} = x_i + \lambda \cdot \sum_{j=1}^k \alpha_j \cdot (x_j - x_i) \quad (4)$$

where:

- x_i denotes a randomly selected minority (ST) sample;
- x_j are the k nearest minority neighbors of x_i in the standardized feature space;
- $\lambda \sim U(0, 1)$ is a scalar randomly drawn from a uniform distribution;
- α_j is the inverse-distance normalized weight of neighbor x_j , defined as:

$$\alpha_j = \frac{1/d(x_i, x_j)}{\sum_{l=1}^k 1/d(x_i, x_l)} \quad (5)$$

- $d(x_i, x_j)$ represents the Euclidean distance between x_i and x_j .

This formulation ensures that the synthetic sample x_{new} lies in the local feature space region occupied by the ST firms and reflects the density-adaptive directionality of the minority class distribution. The generated samples do not contain actual firm codes or time indices; these fields are filled with NaN and NaT to clearly differentiate them from the original observations during subsequent analysis.

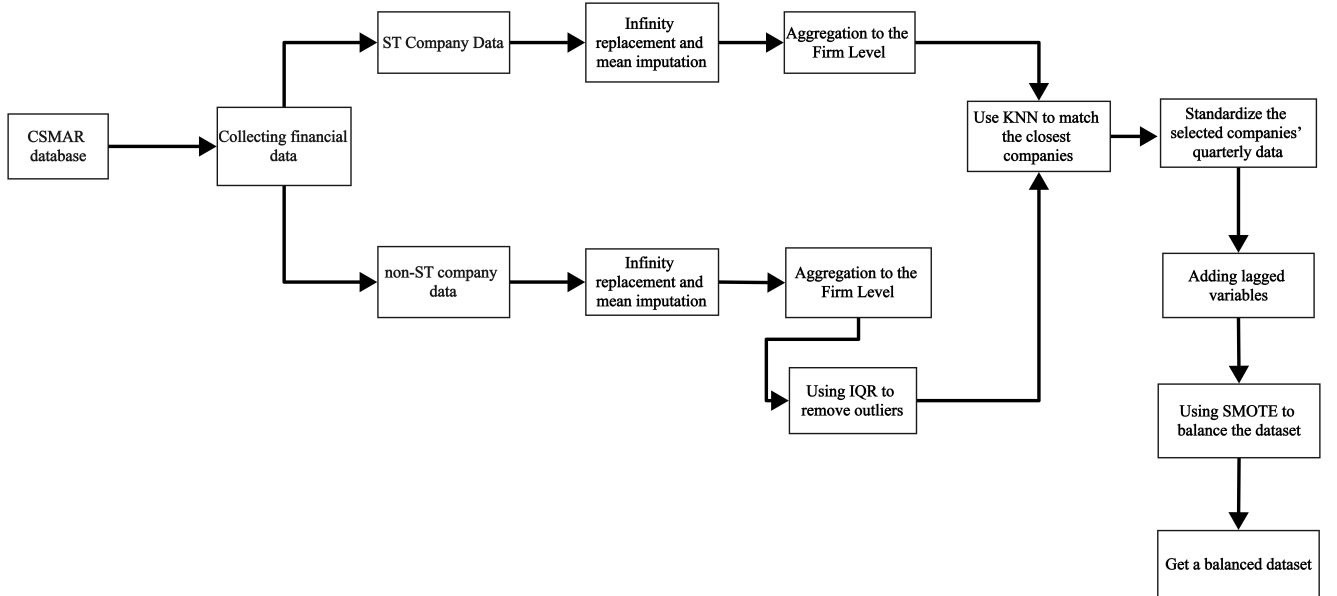


Figure 1: Workflow for Data Cleaning and Preprocessing

4.6 Final Dataset

After all pre-processing steps, a balanced data set was obtained. This dataset contains both original and synthetic samples, with standardized features, lagged variables, and a consistent tabular structure. It forms the foundation for the modeling and analysis stages that follow.

5 Modelling

5.1 Study 1: Static Financial Modelling without Temporal Context

5.1.1 Method

This study provides a baseline for financial distress prediction using only contemporaneous financial ratios. Eleven firm-level indicators were calculated as averages over 20 quarters from 2015Q1 to 2019Q4, covering solvency, profitability, operational efficiency, structure, and growth:

- Solvency: TL/TA , CA/CL , TL/TSE
- Profitability: NP/SR , NP/ASE
- Operational efficiency: SR/ACA , MBC/AI
- Structure and asset quality: FA/TA , SE/FA
- Growth metrics: $TA(t)/TA(t-1)$, $NP(t)/NP(t-1)$

Firms were labeled as $y_i \in \{0, 1\}$ based on ST status in 2019. Given the imbalance (51 ST vs. 5763 non-ST), SMOTE was applied to augment the minority class. The resulting dataset was fed into four classifiers:

$$\text{Logistic Regression, SVM, Random Forest, XGBoost} \quad (6)$$

We evaluated performance using accuracy, F1-score, and AUC (Chen et al., 2015). Feature importance was derived from coefficients (for linear models) and impurity-based scores (for tree models).

5.1.2 Result

Across all four models, XGBoost and Random Forest achieved the best results, with AUC values of 0.9745 and 0.9736 respectively, and F1-scores exceeding 0.91. Logistic Regression, while more interpretable, yielded a significantly lower AUC of 0.7528 and F1-score of 0.68. Analysis of feature importance indicated that profitability metrics—particularly $NP(t)/NP(t-1)$ and NP/SR —consistently ranked highest, suggesting their critical role in static distress classification (Menze et al., 2009). However, both SVM and Logistic Regression exhibited low recall on ST firms, revealing a bias toward non-distressed classifications.

Table 3: Performance of Four Models without Lagged Variables (ST=1 as positive class)

2*Model	2*Accuracy	2*F1 Score	2*AUC Score	Precision			Recall		
				Class 0	Class 1	Macro Avg	Class 0	Class 1	Macro Avg
Logistic Regression	0.7243	0.6796	0.7528	0.68	0.81	0.74	0.86	0.58	0.72
SVM	0.7411	0.6608	0.7777	0.66	0.96	0.81	0.98	0.50	0.74
Random Forest	0.9152	0.9132	0.9736	0.90	0.93	0.92	0.94	0.89	0.92
XGBoost	0.9141	0.9122	0.9745	0.90	0.93	0.91	0.94	0.89	0.91

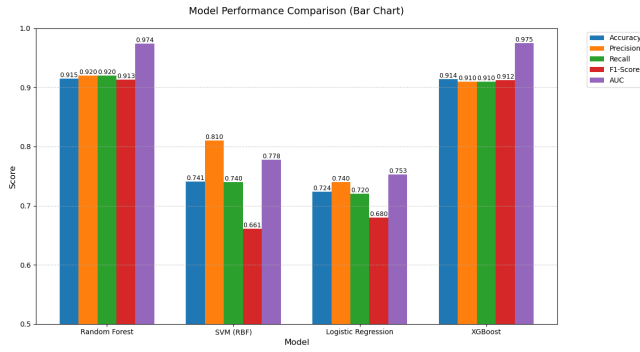


Figure 2: Model Performance Comparison

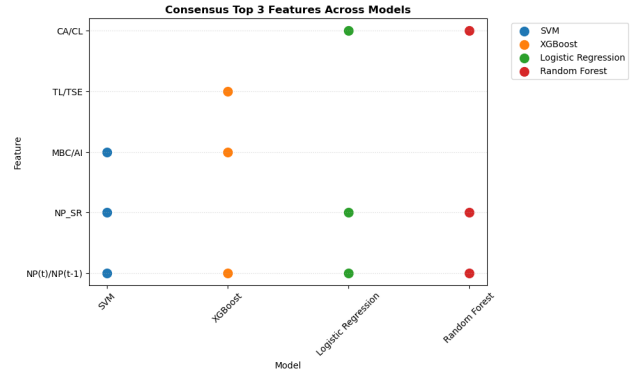


Figure 3: Consensus Top 3 Features Across Models

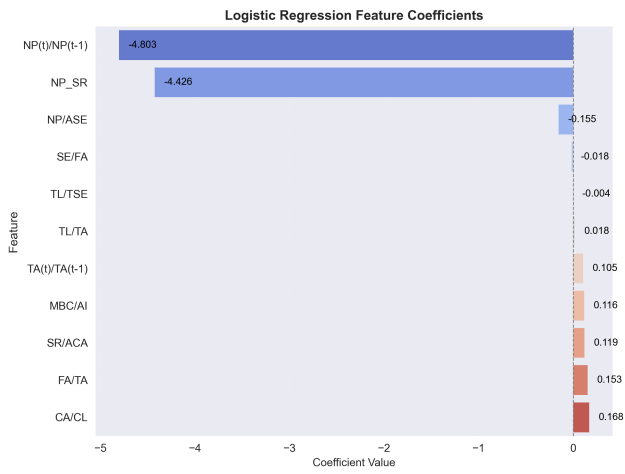


Figure 4: Logistic Regression Feature Coefficients

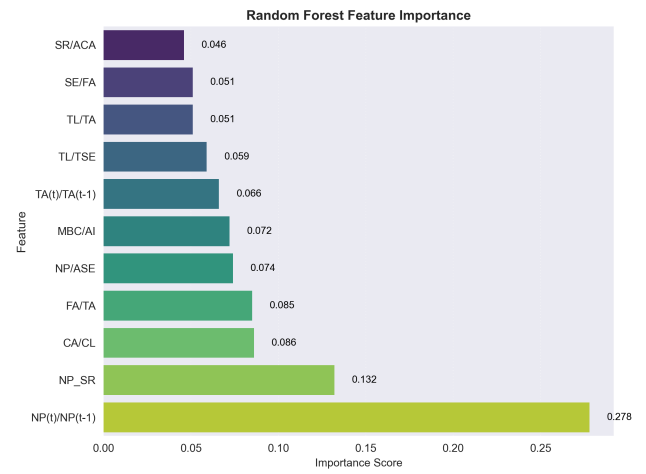


Figure 5: Random Forest Feature Importance

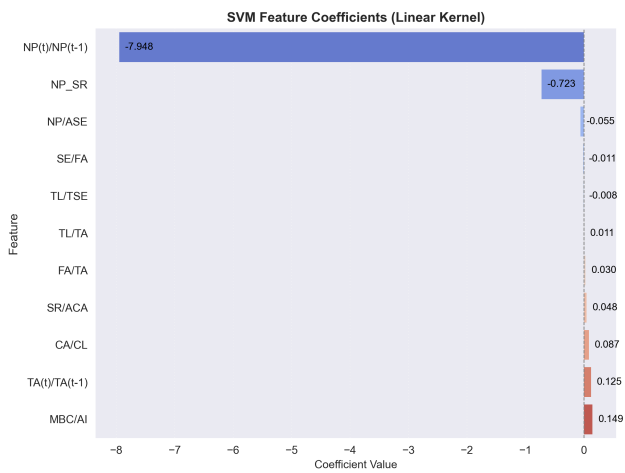


Figure 6: SVM Feature Importance

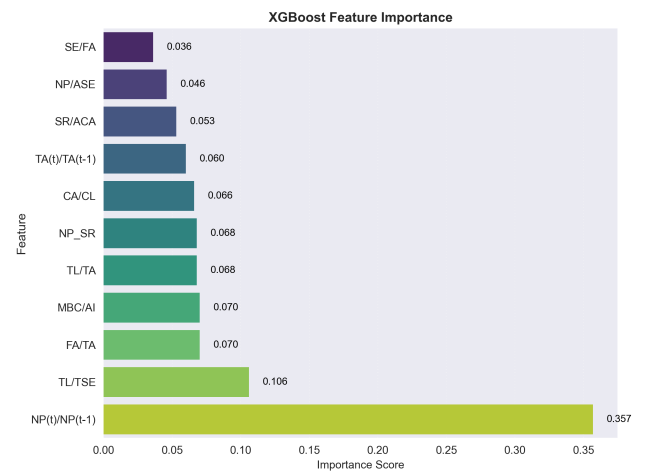


Figure 7: XGB Feature Importance

5.1.3 Discussion

Study 1 demonstrates that static financial ratios can partially distinguish distressed from healthy firms, especially when profitability sharply deviates. Yet the lack of temporal structure limits the model’s ability to capture early-stage financial erosion. Linear models, while interpretable, fail to model non-linear interactions among features. More crucially, this setup does not allow validation of **H1** or **H2**, as no lagged data is present. These limitations emphasize the need for temporal modeling, forming the rationale for Study 2.

5.2 Study 2: Temporal Dynamics through Lagged Financial Indicators

5.2.1 Method

To rigorously test H1 and H2, this study enhances the modeling framework by incorporating lagged financial indicators that reflect firm dynamics over time. Each of the 11 base indicators $x^{(j)}$ was expanded into a temporal sequence covering three time slices: current quarter (t), one-period lag ($t-1$), and two-period lag ($t-2$). This yielded a total of 33 predictor variables per firm:

$$\mathbf{x}_i = \left[x_t^{(1)}, \dots, x_t^{(11)}, x_{t-1}^{(1)}, \dots, x_{t-1}^{(11)}, x_{t-2}^{(1)}, \dots, x_{t-2}^{(11)} \right] \quad (2)$$

The construction of each lagged feature follows a general rule:

$$x_{t-k}^{(j)} = \text{Value of indicator } j \text{ for firm } i \text{ at time } t-k, \quad \text{for } k \in \{1, 2\}, j \in \{1, \dots, 11\} \quad (3)$$

This temporal stacking ensures that models can capture sequential changes in financial performance and detect early-stage deterioration patterns often missed by static snapshots (Li, 2022).

To address class imbalance, we implemented a K-nearest neighbor matching algorithm ($K = 3$) to identify 112 non-ST firms with minimal Euclidean distance to the 51 ST firms in the standardized feature space. This matching ensures the control group is structurally comparable in financial characteristics.

Subsequently, SMOTE was applied on the matched dataset to generate synthetic ST-like samples, further enhancing model robustness under imbalanced conditions. The SMOTE formulation follows:

$$x_{\text{new}} = x_i + \lambda \cdot \sum_{j=1}^k \alpha_j (x_j - x_i), \quad \text{where } \lambda \sim U(0, 1), \quad \sum_{j=1}^k \alpha_j = 1 \quad (4)$$

$$\alpha_j = \frac{1/d(x_i, x_j)}{\sum_{\ell=1}^k 1/d(x_i, x_\ell)} \quad (5)$$

After dataset construction, we retrained all four classifiers—Logistic Regression, Support Vector Machine, Random Forest, and XGBoost—on the extended feature space.

5.2.2 Result

Introducing lagged features led to consistent and notable improvements in model performance across all classifiers. As shown in Table 4, XGBoost’s AUC increased from 0.9745 (static) to 0.9794 (lagged), while Random Forest’s AUC improved from 0.9736 to 0.9840. Notably, both tree-based models achieved substantial gains in F1 Score and accuracy, with Random Forest reaching an accuracy of 0.9343 and XGBoost 0.9170. This suggests enhanced robustness in handling both distressed and non-distressed firms.

Performance gains were also observed in linear and kernel-based models. Logistic Regression improved its AUC from 0.7528 to 0.8156, and SVM from 0.7777 to 0.8254, confirming the effectiveness of temporal dynamics even in simpler classifiers. However, their recall for distressed firms remained relatively low, as shown in the classification reports.

Further analysis reveals that after incorporating lagged features:

- **Precision–Recall Balance Improved:** For both class labels (0: non-ST, 1: ST), Random Forest and XGBoost achieved balanced precision (0.93–0.94) and recall (0.93–0.94).
- **F1 Score Enhancement:** Logistic Regression and SVM improved F1 Scores from 0.68/0.74 to 0.7535 and 0.7356, respectively.
- **Class-Specific Recall Gain:** The recall for ST firms (label 1) increased in XGBoost from 0.85 (static) to 0.90 (lagged), showing enhanced early-risk detection.

Table 4: Performance Comparison Before and After Adding Lagged Variables

Model	AUC (Static)	AUC (Lagged)	F1 (Static)	F1 (Lagged)	Accuracy (Static)	Accuracy (Lagged)
Logistic Regression	0.7528	0.8156	0.6800	0.7535	0.7520	0.7819
SVM	0.7777	0.8254	0.7400	0.7356	0.7630	0.7844
Random Forest	0.9736	0.9840	0.9100	0.9340	0.9120	0.9343
XGBoost	0.9745	0.9794	0.8990	0.9155	0.9030	0.9170

5.2.3 Discussion

The empirical results provide robust evidence in support of both **H1** and **H2**.

Firstly, regarding **H1**, the inclusion of lagged financial indicators significantly improved the predictive accuracy across all models. For instance, the AUC of the XGBoost model increased from 0.9745 to 0.9794, while the Random Forest model saw a rise from 0.9736 to 0.9840. This improvement demonstrates that past financial signals contain valuable information for distinguishing ST firms from healthy ones. In particular, lagged profitability metrics such as $NP(t)/NP(t-1)_{t-2}$ emerged as top contributors, indicating that early-stage financial deterioration manifests in declining profit momentum.

Secondly, with regard to **H2**, the addition of multi-period features—extending from the current quarter x_t to the two preceding quarters x_{t-1} and x_{t-2} —allowed models to better recognize distress patterns as cumulative trajectories rather than one-off abnormalities. This aligns with the notion that financial distress is not a singular shock but the result of sustained erosion in financial performance. The learning function expands from:

$$\hat{y} = f(x_t) \quad \text{to} \quad \hat{y} = f(x_t, x_{t-1}, x_{t-2}), \quad (7)$$

and the latter clearly outperforms the former in both discriminatory power and interpretability:

$$f(x_t, x_{t-1}, x_{t-2}) \gg f(x_t) \quad (8)$$

Moreover, tree-based models benefited more than linear ones, suggesting that non-linear interactions between temporal features better capture complex risk dynamics (Clark and Pregibon, 2017). For example, in Random Forests, SHAP interaction values revealed that the combination of lagged solvency and profitability variables contributed more jointly than individually. In practical terms, this means that regulators and investors should not rely solely on recent snapshots of financial statements. Instead, integrating past financial trajectories offers a more reliable early-warning mechanism. This finding is particularly relevant in the Chinese market context, where ST classification may be triggered by sustained losses or deteriorating net assets over several quarters (Geng, Bose and Chen, 2015). The ability to model and anticipate such conditions can enable earlier intervention and reduce misclassification risks.

In summary, Study 2 validates that lagged financial features are not just supplemental but foundational for robust financial distress modeling, both statistically and practically.

5.3 Study 3: SHAP-Based Interpretation of Temporal Importance

5.3.1 Method

To test **H3**, this study applies SHAP (SHapley Additive exPlanations) values to interpret how each financial indicator—across three time points ($t, t-1, t-2$)—contributes to the prediction of financial distress. SHAP provides a unified framework based on cooperative game theory, decomposing a model’s output into feature-level contributions:

$$\hat{y}(x) = \phi_0 + \sum_{i=1}^{33} \phi_i(x_i) \quad (9)$$

Here, ϕ_0 represents the base value (i.e., the expected model output), and ϕ_i quantifies the marginal impact of the i -th feature x_i on the final prediction for a specific sample. The summation spans all 33 variables, representing 11 financial indicators over 3 time periods.

We selected XGBoost and Random Forest for SHAP analysis due to their superior predictive performance and compatibility with tree-based SHAP algorithms (Marcílio and Eler, 2020). The evaluation focuses on both:

- **SHAP summary plots:** visualize the distribution of SHAP values across samples, revealing which features contribute most (positively or negatively) to classification, along with their value gradients.
- **SHAP bar plots:** depict the average absolute SHAP value per feature, identifying the most influential predictors across the dataset.

To enhance time-disentangled interpretability, each feature was tagged with its lag index (e.g., $NP(t)/NP(t-1)_{lag2}$), allowing us to compare their predictive relevance chronologically. This provides a granular view of which financial signals act as early warning indicators and which serve as reactive markers.

Furthermore, SHAP interaction values were computed using Random Forest to explore potential second-order dependencies between indicators and their temporal counterparts. These interactions help identify compound risk patterns—e.g., how a high lagged profitability ratio amplifies the risk when combined with deteriorating solvency in recent quarters (Lee and Hsieh, 2013).

By aggregating and comparing SHAP scores across $t, t-1$, and $t-2$, we were able to track the persistence and evolution of financial signal strength over time, directly addressing the temporal hypothesis **H3**.

5.3.2 Result

SHAP analysis revealed distinct temporal patterns in feature importance, aligning closely with the hypothesis **H3**. Across both XGBoost and Random Forest, lagged versions of profitability indicators stood out as dominant contributors:

- In XGBoost, the two-period lagged profitability ratio $NP(t)/NP(t-1)_{t-2}$ achieved the highest mean SHAP value (+1.64), followed by $NP(t)/NP(t-1)_{t-1}$ (+1.56), and the contemporaneous value $NP(t)/NP(t-1)_t$ (+1.29).
- SHAP summary plots further illustrated that high values of these lagged features (shown in red) consistently pushed predictions toward the ST class.
- In contrast, models like SVM and Logistic Regression showed lower temporal resolution. Their SHAP distributions were more concentrated around a narrow band near zero, with limited spread, indicating lower explanatory power.

Notably, features such as CA/CL_{t-2} and $TA(t)/TA(t-1)_{t-2}$ also appeared in XGBoost’s top 10 influential variables, suggesting that solvency and growth indicators from earlier quarters provided meaningful foresight into future distress.

Random Forest interaction SHAP values further indicated synergistic effects between $NP(t)/NP(t-1)$ and other lagged features (e.g., FA/TA , MBC/AI), reinforcing that multi-dimensional financial decline can manifest through nonlinear temporal interactions.

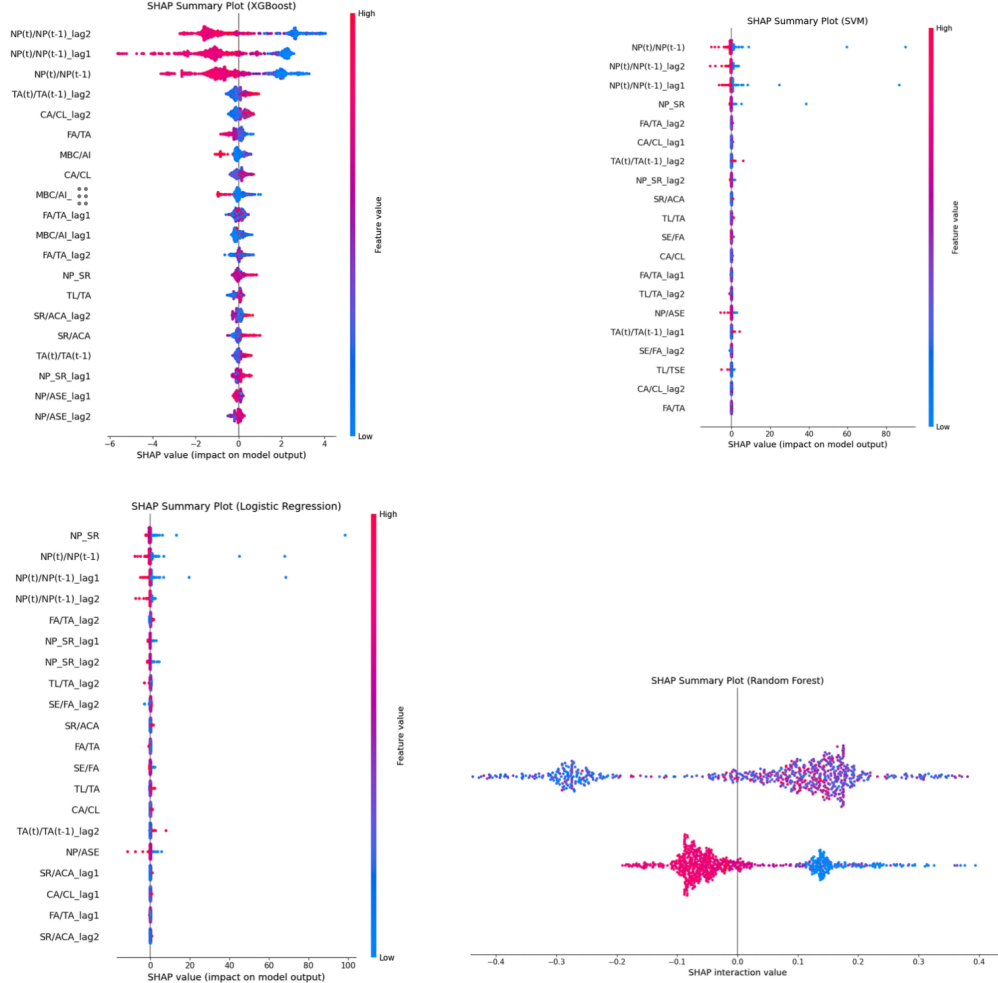


Figure 8: SHAP Summary Plot

5.3.3 Discussion

Study 3 confirms the hypothesis **H3**: SHAP-based interpretation reveals how the relevance of financial indicators evolves over time and provides transparency into the temporal structure of predictive models.

Three main insights emerge:

1. **Temporal asymmetry in signal strength:** Lagged variables—especially those from $t-2$ —consistently outperform their $t-1$ and t counterparts. This supports the assertion that financial deterioration is gradual, and early warning signs reside in trailing indicators.
2. **Model differentiation in interpretability:** Tree-based models such as XGBoost and Random Forest yield richer and more stable SHAP distributions, enabling clearer decomposition of temporal effects. Linear models, despite being interpretable in theory, offer less robust insight due to their parametric constraints and poorer fit to nonlinear patterns.

3. **Regulatory and investor implications:** From a practical standpoint, the dominance of $\phi_{t-2} > \phi_{t-1} > \phi_t$ in SHAP rankings suggests that financial monitoring frameworks should incorporate lagged metrics explicitly. For instance, regulatory stress tests could place higher weights on historical profitability degradation, even when current ratios appear temporarily stable.

In summary, SHAP not only enhances transparency in black-box models but also enables a temporally disentangled view of financial distress. By exposing the structural lag in how financial signals impact predictions, Study 3 provides empirical backing for the design of more proactive early warning systems.

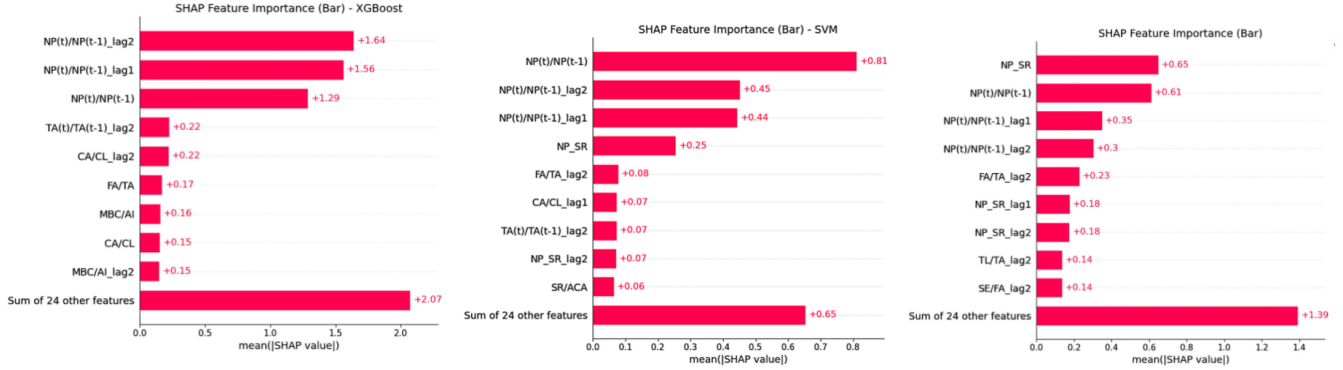


Figure 9: SHAP Feature Importance

6 Conclusion

This empirical investigation presents a multi-stage framework for forecasting financial distress (ST classification) in the Chinese A-share market. Through a progressive modeling design spanning static, temporal, and interpretable dimensions, we demonstrate the critical role of lagged financial indicators in capturing early warning signals.

Study 1 establishes a performance benchmark using contemporaneous financial ratios, revealing the limitations of static modeling in isolating the onset of distress. Study 2 expands upon this by introducing lagged variables from $t-1$ and $t-2$, validating **H1** and **H2** through significantly improved model performance, particularly for tree-based classifiers like XGBoost (AUC = 0.9794) and Random Forest (AUC = 0.9840). The temporal structure enables models to better differentiate short-term volatility from cumulative decline.

Study 3 addresses the transparency imperative by applying SHAP (SHapley Additive exPlanations) to decompose model outputs across time. The results confirm **H3**, showing that two-period lagged profitability ratios ($NP(t)/NP(t-1)_{t-2}$) consistently exhibit the highest contribution scores across models. SHAP further illustrates that tree-based models offer more robust temporal interpretability, making them suitable for regulatory applications and risk management.

Collectively, this research underscores that financial distress is best understood as a dynamic process, where temporal memory embedded in historical indicators yields both superior predictive accuracy and richer interpretive value. These findings advocate for the integration of lag-aware modeling pipelines in corporate risk assessment systems and highlight the potential of explainable machine learning in building trustworthy financial AI tools.

7 Limitations and Future Work

Despite the promising results, this study has several notable limitations.

First, the financial indicators employed are entirely accounting-based and quantitative in nature. Non-financial variables—such as corporate governance structures, industry sentiment, or macroeconomic shocks—were not incorporated. This omission may limit the model’s generalizability, particularly in policy-sensitive or volatile sectors. Future research could enrich the feature space by integrating alternative data sources, including textual disclosures, ESG indicators, and managerial quality proxies.

Second, although the incorporation of lagged variables improved predictive accuracy, the temporal window was limited to three discrete points: t , $t-1$, and $t-2$. No formal temporal modeling was implemented. Sequential models such as Long Short-Term Memory (LSTM) networks or Temporal Fusion Transformers (TFT) may capture deeper memory effects and nonlinear deterioration trends. Future work could explore these approaches while balancing interpretability concerns.

Third, this study focused primarily on cross-sectional interpretability using SHAP values. However, it did not assess the temporal evolution of feature importance for individual firms. Extending SHAP analysis into time-series or applying counterfactual explanation techniques may help establish firm-specific early warning trajectories and allow dynamic monitoring.

Finally, although KNN matching and SMOTE were applied to alleviate class imbalance, synthetic samples inevitably deviate from real-world financial behaviors. As more labeled data becomes available, future research should validate model robustness across multiple years, sectors, and economic cycles to enhance both external validity and practical relevance.

(Word count:2997)

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Appendix

Coding for this project(Using Github to store)

<https://github.com/Ernest-Lin/IOM209-Business-Intelligence-Individual-Coursework.git>

Introduction about database

We collect quarterly financial statement data of all A-share listed companies in 2019 from the CSMAR database . The data spans from 2015Q1 to 2019Q4 and includes firm-level accounting indicators necessary for constructing financial distress prediction models.

Definition of Financial Indicators

- **TL/TA** = Debt Ratio, defined as Total Liabilities divided by Total Assets.
- **CA/CL** = Current Ratio, defined as Current Assets divided by Current Liabilities.
- **TL/TSE** = Leverage Ratio, defined as Total Liabilities divided by Total Shareholders' Equity.
- **NP/SR** = Net Profit Margin, defined as Net Profit divided by Sales Revenue.
- **NP/ASE** = Return on Equity (ROE), defined as Net Profit divided by Average Shareholders' Equity.
- **SR/ACA** = Current Asset Turnover, defined as Sales Revenue divided by Average Current Assets.
- **MBC/AI** = Inventory Turnover, defined as Main Business Cost divided by Average Inventory.
- **TA(t)/TA(t-1)** = Asset Growth Rate, defined as the ratio of current Total Assets to prior-period Total Assets.
- **NP(t)/NP(t-1)** = Profit Growth Rate, defined as the ratio of current Net Profit to prior-period Net Profit.
- **FA/TA** = Fixed Asset Ratio, defined as Fixed Assets divided by Total Assets.
- **SE/FA** = Equity-to-Fixed Asset Ratio, defined as Shareholders' Equity divided by Fixed Assets.

Feature Importance of Logistic Regression (Without Lagged Variables)

Table 5: Logistic Regression Coefficients Ranked by Absolute Value

Feature	Coefficient
NP(t)/NP(t-1)	-4.803
NP_SR	-4.426
CA/CL	0.168
NP/ASE	-0.155
FA/TA	0.153
SR/ACA	0.119
MBC/AI	0.116
TA(t)/TA(t-1)	0.105
SE/FA	-0.018
TL/TA	0.018
TL/TSE	-0.004

Feature Importance of SVM (Without Lagged Variables)

Table 6: SVM Linear Kernel Coefficients Ranked by Absolute Value

Feature	Coefficient
NP(t)/NP(t-1)	-7.948
NP_SR	-0.723
MBC/AI	0.149
TA(t)/TA(t-1)	0.125
CA/CL	0.087
NP/ASE	-0.055
SR/ACA	0.048
FA/TA	0.030
SE/FA	-0.011
TL/TA	0.011
TL/TSE	-0.008

Feature Importance of XGBoost (Without Lagged Variables)

Table 7: XGBoost Feature Importance Ranked by Gain

Feature	Importance Score
NP(t)/NP(t-1)	0.3567
TL/TSE	0.1059
MBC/AI	0.0702
FA/TA	0.0696
NP_SR	0.0682
TL/TA	0.0681
CA/CL	0.0661
TA(t)/TA(t-1)	0.0596
SR/ACA	0.0529
NP/ASE	0.0464
SE/FA	0.0364

Feature Importance of Random Forest (Without Lagged Variables)

Table 8: Random Forest Feature Importance Ranked by Gini Impurity

Feature	Importance Score
NP(t)/NP(t-1)	0.2789
NP_SR	0.1328
CA/CL	0.0867
FA/TA	0.0836
NP/ASE	0.0743
MBC/AI	0.0725
TA(t)/TA(t-1)	0.0659
TL/TSE	0.0580
SE/FA	0.0510
TL/TA	0.0505
SR/ACA	0.0459

Optimal Hyperparameters (Without Lagged Variables)

Before incorporating lagged variables, each model underwent hyperparameter tuning using grid search with cross-validation. The best-performing parameter combinations for the baseline models were as follows:

- **Random Forest:** The model achieved optimal performance with 150 trees, that is, `n_estimators = 150`, using the square root of the number of features at each split, `max_features = 'sqrt'`. It also used the default values for minimum samples to split and leaf size, specifically `min_samples_split = 2` and `min_samples_leaf = 1`. Bootstrapping was disabled with `bootstrap = False`, and no maximum depth was imposed, i.e., `max_depth = None`.
- **Support Vector Machine (SVM):** The optimal configuration employed a linear kernel, `kernel = 'linear'`, with a regularization parameter of `C = 10`, balancing the trade-off between margin width and classification accuracy.
- **XGBoost:** The model performed best using 250 trees, `n_estimators = 250`, with a maximum depth of 8 per tree, `max_depth = 8`. It applied a learning rate of 0.1, full subsampling and column sampling with `subsample = 1.0` and `colsample_bytree = 1.0`. Regularization terms were disabled, with `reg_alpha = 0` and `reg_lambda = 0`. The gamma parameter, which controls the minimum loss reduction required to make a further partition, was set to `gamma = 0`.